

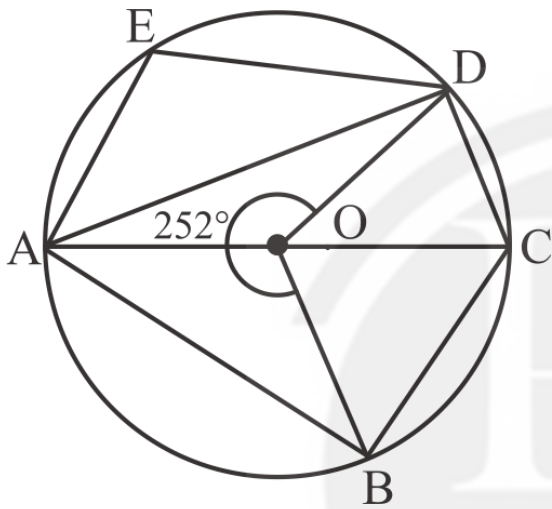
MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

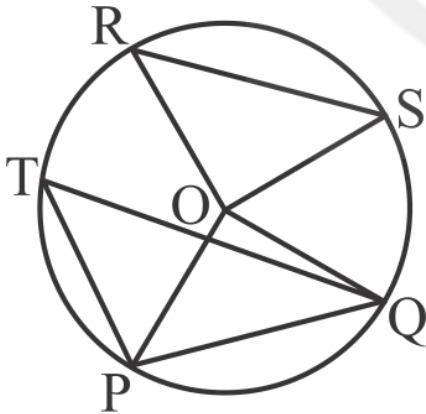
DPP: 2

2D Geometry - 2

- Q1** In the following figure, AC is a diameter of a circle with centre O, which bisects the angle BCD. If $DE \parallel AC$, then find the measure of the angle EDA (in degrees).



- Q2** In the following figure, O is the centre of the circle and $PQ = RS$. If $\angle TPO = 44^\circ$ and $\angle ROS = 64^\circ$, then find the measure of $\angle TQP$ (in degrees).



- Q3** Two concentric circles have radii R and r ($R > r$). A chord of the larger circle is tangent to the smaller circle. If the length of this chord is 40 cm, what is the area of the region between the two circles (in cm^2)?
- (A) 120π (B) 200π

(C) 320π (D) 400π

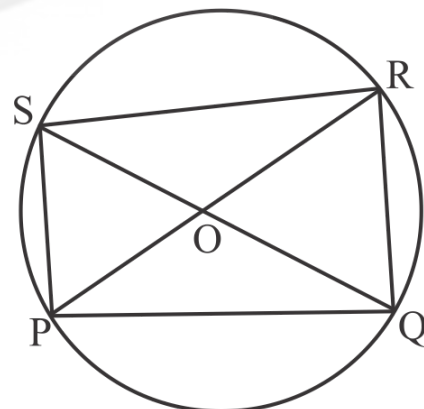
- Q4** The area of the triangle formed by an angle 45° subtended by a chord at the centre of a circle is $18\sqrt{2} \text{ cm}^2$. Find the area of the triangle (in cm^2) formed by an angle 60° subtended by a chord at the centre of the same circle.

(A) $15\sqrt{3}$ (B) $16\sqrt{2}$
(C) $18\sqrt{3}$ (D) $20\sqrt{2}$

- Q5** Two circles of radii 25 cm and 39 cm intersect at the points A and B. If $AB = 30$ cm and then find the distance between the centres of the circles (in cm).

(A) 40 (B) 56
(C) 64 (D) 72

- Q6** A quadrilateral PQRS is inscribed in a circle such that $PQ : SR = 6 : 5$ and $SP : RQ = 3 : 10$. If PR and QS intersect each other at O, then find the ratio $SO : OQ$.



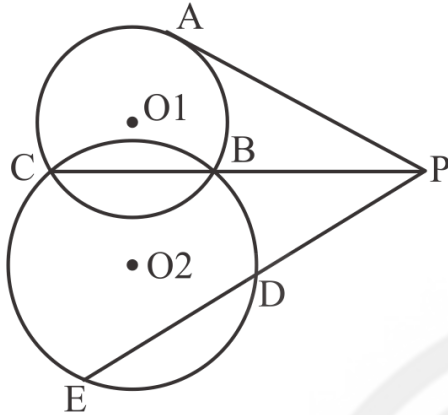
(A) 1 : 4 (B) 4 : 9
(C) 3 : 10 (D) 2 : 5

- Q7** Two chords AB and PQ intersect at point O inside a circle, dividing AB in the ratio 2:3. If



the lengths of AB and PQ are in the ratio 2:5, and AB is 10 cm, find the sum of all possible lengths of PO (in cm).

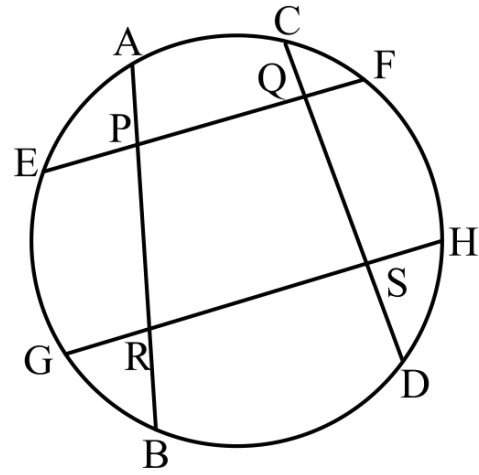
- Q8** In the following figure, O_1 and O_2 are the centres of two intersecting circles such that $PA = 18$ cm and $PE = 24$ cm. Then, find the length of the chord DE (in cm).



- (A) 9.5 (B) 10
(C) 10.5 (D) 11

- Q9** Let P be a point outside an O-centric circle and A, B, C are the points on the circle such that PA is tangent, PBC is a secant and D is the mid-point of BC. If $PB \cdot PC = 144$ cm, $PO = 13$ cm, and $OD = 3$ cm, then find the area of the triangle OBC (in cm^2).

- Q10** 4 chords AB, CD, EF and GH of a circle intersect each other within the circle as shown in the figure, forming a quadrilateral PQSR such that each intersecting pair of chords divide each other in the ratio of 1: 5. If the lengths of the pair of chords whose parts form opposite sides of the quadrilateral are 24 cm, 18 cm; and 12 cm, 30 cm respectively, then find the perimeter (in cm) of the quadrilateral PQSR.



- (A) 50 (B) 52
(C) 54 (D) 56

- Q11** A is the center of a circle, and B is a point outside the circle. BC is a tangent to the circle at C, with C being the point of contact. The line connecting B and A intersects the circle at D (located between B and A). The tangent to the circle at D intersects BC at E. Given that $BD = 24$ cm and $BC = 36$ cm, determine the length of AE?

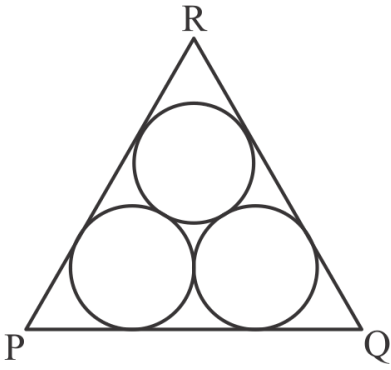
- (A) $5\sqrt{13}$ cm (B) $10\sqrt{3}$ cm
(C) 15 cm (D) $10\sqrt{5}$ cm

- Q12** Two circles intersect at points X and Y. Point Q is situated outside both the circles such that Y-X-Q forms a straight line. The line QR touches the circle centered at O_1 as a tangent. The extended chord FG of the second circle, centered at O_2 , passes through point Q. Given that $QR = 12$ cm and $QG = 8$ cm, determine the length of chord FG.

- (A) 8 cm (B) 9 cm
(C) 10 cm (D) 12 cm

- Q13** Three equal circles have been inscribed in an equilateral triangle PQR as shown below. The area of PQR is $(32+16\sqrt{3}) \text{ cm}^2$. What will be the area of triangle formed by joining the centers of the three circles in square centimeters?





- Q14** In a circle, four points P, Q, R and S are on the circumference of the circle such that P and Q are as far as possible. The angle PQR is 60° . The length of QR is 25 cm and length of QS is 14 cm then find the area of triangle PQS in cm^2 .

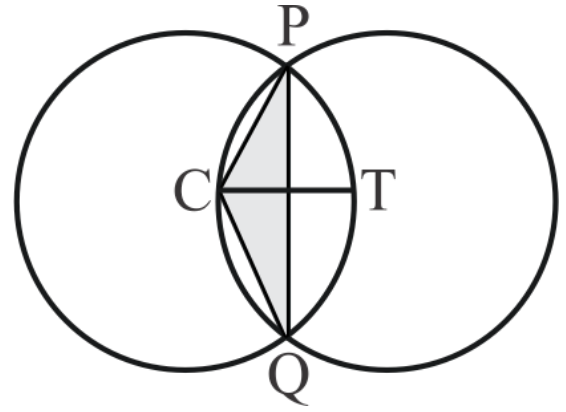
(A) 441 (B) 366
(C) 315 (D) 336

- Q15** A sector of a circle has a central angle of 20° of the complete angle and an arc length of 44 cm. Find the radius (in cm) of the circle. [Take $\pi = \frac{22}{7}$]

- Q16** An O-centric circle is inscribed inside a rectangle ABCD such that the circle touches three sides AB, BC and CD of the rectangle. If the area of the circle is $16\pi \text{ cm}^2$, then find the perimeter of the triangle OBC (in cm).

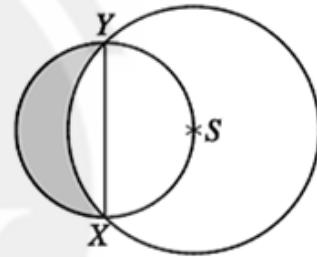
(A) $8(2 + \sqrt{2})$ (B) $2\sqrt{2}(2 + \sqrt{2})$
(C) $4(1 + \sqrt{2})$ (D) $8(1 + \sqrt{2})$

- Q17** Find the area of the shaded region (in cm^2) provided that C, T are the centres of the circles and the area of the circle with centre C is $144\pi \text{ cm}^2$.



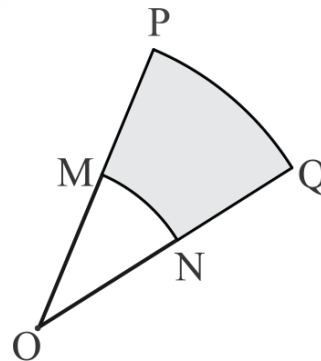
(A) $18\sqrt{3}$ (B) $24\sqrt{3}$
(C) $32\sqrt{3}$ (D) $36\sqrt{3}$

- Q18** If in the following figure, S is the centre of the larger circle, and the area of the shaded region is 8 cm^2 , what is the area of the larger circle (in cm^2) if XY passes through the centre of the smaller circle?



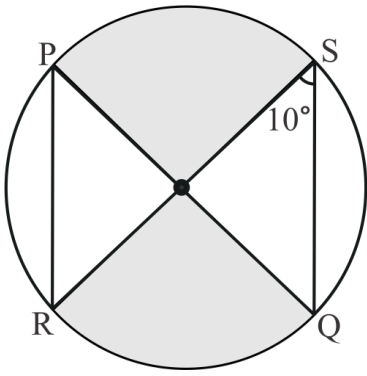
(A) 8π (B) 12π
(C) 16π (D) 20π

- Q19** In the following figure, the perimeter of the shaded region is 20 cm, $ON = 3 \text{ cm}$, and QN has the maximum possible integer value. Find the area of the shaded region (in cm^2).

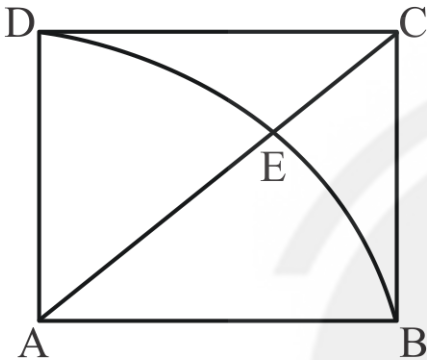


- Q20** If PQ and RS are diameters of circle and radius is 42 cm. Find the area of the shaded region (in sq. cm) if $\angle RSQ = 10^\circ$. [Take $\pi = \frac{22}{7}$]

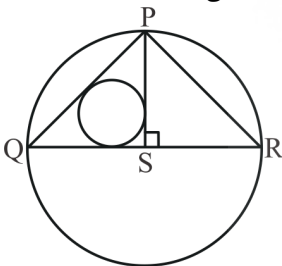




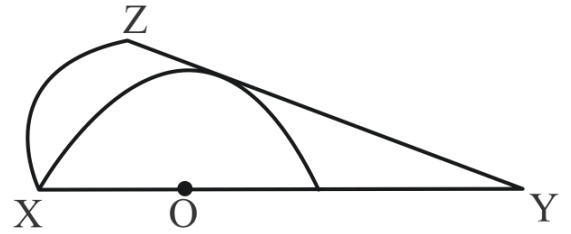
- Q21** In the following diagram, ABCD is a square. Find the length of EC (in cm) if area of the A-centric-circle is $(3 + 2\sqrt{2})\pi \text{ cm}^2$.



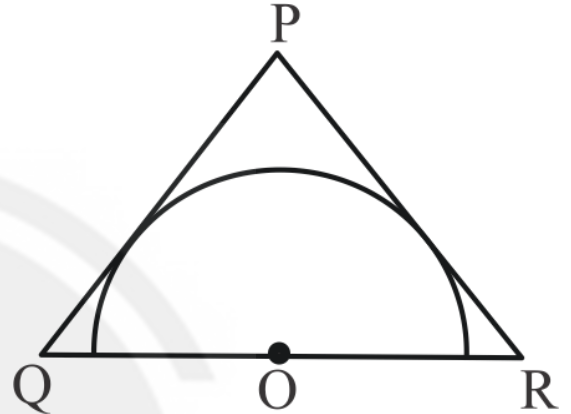
- Q22** In the following figure, PQ = 30 cm, PR = 40 cm, and PS is a perpendicular line drawn from P to the side QR, the diameter of the circle. Find the difference between the areas of the smaller and larger circles.



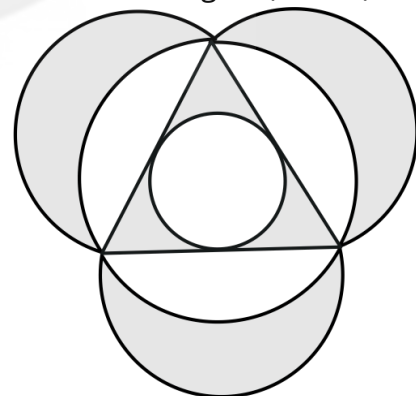
- (A) $\frac{589\pi}{4}$ (B) $\frac{589\pi}{2}$
 (C) 589π (D) None of these
- Q23** A semicircle of area $\frac{9\pi}{8} \text{ cm}^2$ is inscribed inside a sector XYZ of a circle (as shown in the figure) with centre O. If YZ = 6 cm, find the length of the perpendicular line drawn from Z to XY (in cm).



- Q24** In a triangle PQR, a semicircle with centre O is inscribed (as shown in the figure). If PQ = 16 cm, QR = 34 cm and PR = 30 cm, find the ratio of the area of the semicircle to that of the triangle.



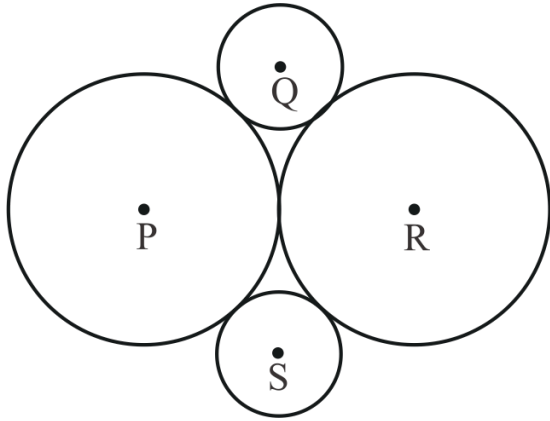
- (A) $280\pi : 529$
 (B) $240\pi : 529$
 (C) $160\pi : 529$
 (D) $120\pi : 529$
- Q25** The triangle shown in the following figure is an equilateral triangle with a side length of 12 cm. Taking the sides of the triangle as diameters, 3 semicircles are drawn. Then, find the area of the shaded region (in cm^2).



- (A) $90\sqrt{3} - 9\pi$ (B) $72\sqrt{3} - 6\pi$
 (C) $54\sqrt{3} - \pi$ (D) $18\sqrt{3} - \frac{\pi}{3}$
- Q26** Four circles with centres P, Q, R and S are arranged such that Quadrilateral PQRS is a



square. If circles with centres Q and S are of radius of length 2 cm, then what is the area of the quadrilateral PQRS?

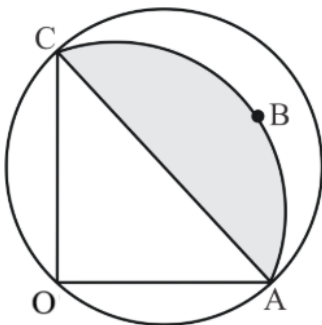


- (A) $12(2 + \sqrt{2})$ (B) $8(3 + 2\sqrt{2})$
 (C) $4(3 + \sqrt{2})$ (D) $2(3 + 2\sqrt{2})$

Q27 A circle is cut out from a rectangle of area 81 cm^2 in such a way that the circle touches two opposite sides of the rectangle. If the area of the cut-out portion is $\frac{1}{3}$ rd of the leftover portion, then find the perimeter of the rectangle (in cm).

- (A) $2 \times 9\sqrt{\pi}(1 + \frac{1}{\pi})$
 (B) $\frac{9\sqrt{\pi}}{4}(1 + \frac{3}{\pi})$
 (C) $\frac{9\sqrt{\pi}}{8}(1 + \frac{5}{\pi})$
 (D) $9\sqrt{\pi}(1 + \frac{1}{2\pi})$

Q28 From a circle of radius $14\sqrt{2} \text{ cm}$, if the shaded portion is cut out (as shown in the figure), then find the area of the leftover portion of the circle (in cm^2), provided that OABCO is a quadrant of a circle.



- (A) 1080 (B) 1200
 (C) 1008 (D) 1012

Q29 A circle with center O has chords PQ measuring 18 cm and RS measuring 12 cm. When extended beyond points Q and S, respectively, PQ and RS intersect at point T, located outside the circle. If $\angle PTS$ is 60° and QT is 14 cm, determine the area of the circle.

- (A) $207\pi \text{ cm}^2$
 (B) $225\pi \text{ cm}^2$
 (C) $218\pi \text{ cm}^2$
 (D) $228\pi \text{ cm}^2$

Q30 Three identical circles, each with a radius of 3.5 units, are arranged so that each circle touches the other two at exactly one point. Tangents are drawn between each pair of circles that do not intersect the third circle. These three tangents form a triangle. What is the area of the triangle that is not covered by the circles?

- (A) $(56\sqrt{3} - 63) \text{ cm}^2$
 (B) $(56\sqrt{3} - 42) \text{ cm}^2$
 (C) $(49\sqrt{3} - 63) \text{ cm}^2$
 (D) $(49\sqrt{3} - 42) \text{ cm}^2$



Answer Key

Q1 27
Q2 46
Q3 D
Q4 C
Q5 B
Q6 A
Q7 25
Q8 C
Q9 12
Q10 D
Q11 A
Q12 C
Q13 8
Q14 D
Q15 35

Q16 D
Q17 D
Q18 C
Q19 9
Q20 616
Q21 1
Q22 C
Q23 2
Q24 D
Q25 B
Q26 B
Q27 A
Q28 C
Q29 D
Q30 D



Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Since, $DE \parallel AC$, so $\angle EDA = \angle DAC$.

Now, given that reflex $\angle DOB = 252^\circ$, and so $\angle DCB = \frac{\text{reflex } \angle DOB}{2} = \frac{252^\circ}{2} = 126^\circ$ (since $\angle DOB$ forms a diamond figure, i.e., since the angle subtended by an arc at the center is twice the angle subtended at the circumference.)

Therefore, $\angle DCA = \frac{\angle DCB}{2} = \frac{126^\circ}{2} = 63^\circ$

[Since the diameter AC bisects the angle $\angle DCB$]

Thus, from the triangle DAC , we get

$\angle DAC = 180^\circ - \angle ADC - \angle DCA = 180^\circ - 90^\circ - 63^\circ = 27^\circ$ [Since, $\angle ADC$ is an angle in a semi-circle.]

Video Solution:



Q2 Text Solution:

Since the lengths of the chords RS and PQ are equal, so the angles subtended by them at the centre of the circle are equal.

Therefore, $\angle ROS = \angle QOP = 64^\circ$

Now, $\angle PTQ = \frac{\angle QOP}{2} = \frac{64^\circ}{2} = 32^\circ$ [Angle which is subtended by an arc at the center of a circle is double the angle subtended at any point on the circumference]

$\angle OPQ = \angle OQP$ [since, $OP = OQ = \text{Radius}$]

So, in $\triangle OPQ$,

$\angle OPQ + \angle OQP + \angle POQ = 180^\circ$

$2 \angle OPQ + 64^\circ = 180^\circ$

$\angle OPQ = 58^\circ$

Therefore, $\angle TPQ = \angle TPO + \angle OPQ = 44^\circ + 58^\circ = 102^\circ$

Thus, in $\triangle TPQ$,

$\angle TPQ + \angle TQP + \angle PTQ = 180^\circ$

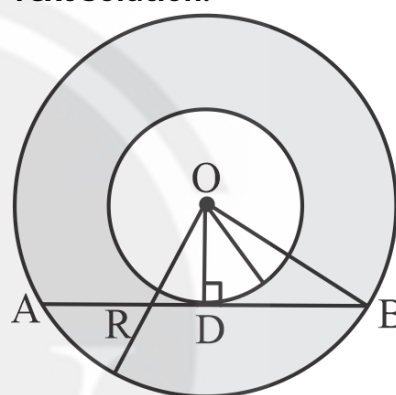
$102^\circ + \angle TQP + 32^\circ = 180^\circ$

$\angle PQR = 46^\circ$

Video Solution:



Q3 Text Solution:



Let the chord length $AB = 40$ cm.

Half the chord length: $BD = \frac{AB}{2} = 20$ cm.

Using the Pythagorean theorem for the triangle OBD :

$$\begin{aligned} r^2 + BD^2 &= R^2 \implies r^2 + 20^2 = R^2 \\ &\implies R^2 - r^2 = 400. \end{aligned}$$

Area of the region between the two circles:

$$\pi(R^2 - r^2) = \pi \times 400 = 400\pi \text{ cm}^2$$

Video Solution:



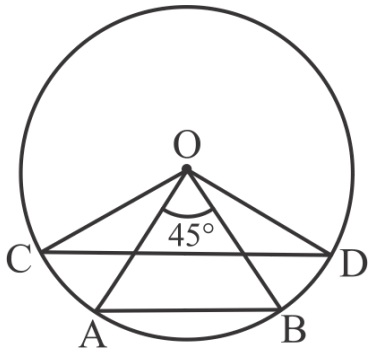
Q4 Text Solution:



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According to the given data, the above diagram is drawn. Now, since the area of the triangle $OAB = 18\sqrt{2} \text{ cm}^2$, therefore, we get

$$\frac{1}{2} \times OA \times OB \times \sin 45^\circ = 18\sqrt{2}$$

$$\frac{1}{2} \times OA^2 \times \frac{1}{\sqrt{2}} = 18\sqrt{2}$$

[Since, $OA = OB = \text{Radius}$]

$$OA = 6\sqrt{2} \text{ cm}$$

i.e., the radius of the circle is $6\sqrt{2} \text{ cm}$.

Now, the area of the required triangle OCD =

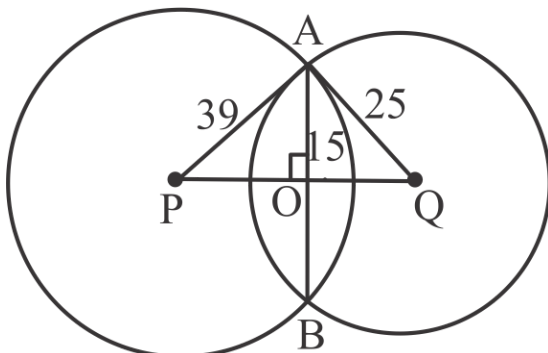
$$\frac{1}{2} \times OC \times OD \times \sin 60^\circ = \frac{1}{2} \times (6\sqrt{2})^2$$

$$\times \frac{\sqrt{3}}{2} = 18\sqrt{3} \text{ cm}^2$$

Video Solution:



Q5 Text Solution:



Since the common chord is perpendicularly bisected by the distance between the centres of two intersecting circles, so $AO = OB =$

$$\frac{30}{2} = 15 \text{ cm.}$$

Now, in the right-angled triangle APO , $AP = 39 \text{ cm}$, $AO = 15 \text{ cm}$, so by using Pythagorean Theorem, we get

$$PO = \sqrt{AP^2 - AO^2} = \sqrt{39^2 - 15^2} \text{ cm.} = 36$$

Also, in the right-angled triangle AQO , $AQ = 25 \text{ cm}$, $AO = 15 \text{ cm}$, so we have

$$OQ = \sqrt{AQ^2 - OA^2} = \sqrt{25^2 - 15^2} \text{ cm.} = 20$$

Thus, the required distance between the centres of the circles = $PQ = PO + OQ = 36 + 20 = 56 \text{ cm}$.

Video Solution:



Q6 Text Solution:

In the given problem,

$\angle POQ = \angle SOR$ [Opposite angles],

$\angle OQP = \angle ORS$, and $\angle QPO = \angle RSO$

subtended by the same arc at the circumference of a circle are equal,]

So, $\triangle POQ \sim \triangle SOR$

$$\text{Thus, } \frac{PO}{SO} = \frac{OQ}{OR} = \frac{PQ}{SR} = \frac{6}{5} \dots(i)$$

Similarly, $\triangle SOP \sim \triangle ROQ$

$$\text{Thus, } \frac{SO}{RO} = \frac{OP}{OQ} = \frac{SP}{RQ} = \frac{3}{10} \dots(ii)$$

So, we have

$$\frac{SO}{OQ} = \frac{SO}{RO} \times \frac{RO}{OQ} = \frac{3}{10} \times \frac{5}{6} = \frac{1}{4}$$

Video Solution:



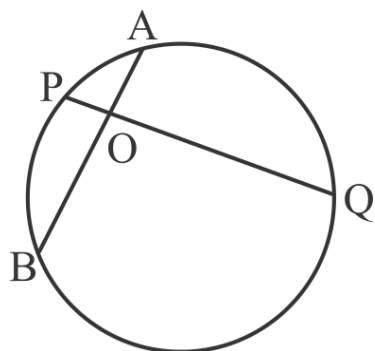
Q7 Text Solution:



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$$AB : PQ = 2 : 5$$

$$AO : OB = 2 : 3$$

$$AB = 10 \text{ cm}$$

$$\text{Therefore, } PQ = 5 \times 5 = 25 \text{ cm}$$

$$\text{So, } AO = 10 \times \frac{2}{5} = 4 \text{ cm, and}$$

$$OB = 10 - 4 = 6 \text{ cm}$$

Now, we know that for any two intersecting chords AB and PQ at O,

$$AO \times OB = PO \times OQ$$

$$\text{i.e., } PO \times OQ = 4 \times 6 = 24 \text{ cm ... (i)}$$

$$\text{Again, } PQ = 25 \text{ cm}$$

$$\text{i.e., } PO + OQ = 25 \text{ ... (ii)}$$

(i) and (ii) implies,

$$PO \times (25 - PO) = 24$$

$$25PO - PO^2 - 24 = 0$$

$$PO^2 - 25PO + 24 = 0$$

$$PO = 24 \text{ cm or } 1 \text{ cm.}$$

$$\text{So, the sum of the possible lengths of } PO = 24 + 1 = 25 \text{ cm}$$

Video Solution:



Q8 Text Solution:

Using the Power of a Point Theorem, we get for the circle with centre O1:

$$PA^2 = PB \times PC \text{ and}$$

for the circle with centre O2:

$$PB \times PC = PD \times PE$$

Thus, combining the data, we get

$$PA^2 = PD \times PE$$

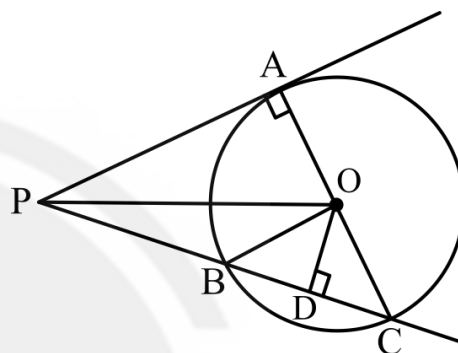
$$PD = \frac{PA^2}{PE} = \frac{18^2}{24} = 13.5 \text{ cm}$$

$$\text{Thus, } DE = PE - PD = 24 - 13.5 = 10.5 \text{ cm}$$

Video Solution:



Q9 Text Solution:



We know that, If P is a point outside a circle and A, B, C are points on the circle such that PA is a tangent and

PBC is a secant then $PA^2 = PB \times PC$.

Given that, $PB \times PC = 144$

So, $PA = 12$ (since, length cannot be negative)

Also, given that, $PO = 13 \text{ cm}$

So, from the right-angled triangle AOP, we have

$$AO = \sqrt{PO^2 - PA^2} = \sqrt{13^2 - 12^2} = 5 \text{ cm}$$

i.e., the radius of the circle = 5 cm.

Now, since, D is the mid-point of BC, let us join the line OD which will also be the perpendicular line on BC from O. Therefore, OD is the height of the right-angled triangle ODC.

Again, $OD = 3 \text{ cm}$ is given.

Therefore,

$$DC = \sqrt{OC^2 - OD^2} = \sqrt{5^2 - 3^2} = 4 \text{ cm.}$$

$$\text{Hence, } BC = 2DC = 2 \times 4 = 8 \text{ cm}$$



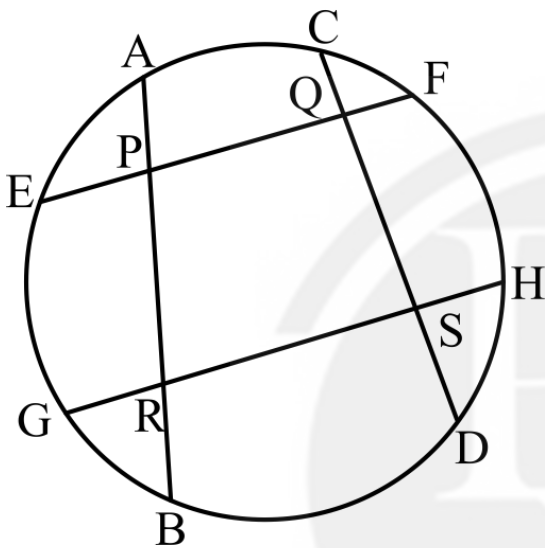
Thus, the area of the triangle OBC =

$$\frac{1}{2} \times BC \times OD = \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$$

Video Solution:



Q10 Text Solution:



So, according to the question, we have

$$CQ : QD = 1 : 5$$

$$FQ : QE = 1 : 5$$

$$AP : PB = 1 : 5$$

$$EP : PF = 1 : 5$$

$$GR : RH = 1 : 5$$

$$BR : AR = 1 : 5$$

$$DS : SC = 1 : 5$$

$$HS : SG = 1 : 5$$

$$\text{Also, } PR = AB - AP - BR = 24 -$$

$$\frac{1}{6} \times 24 - \frac{1}{6} \times 24 = 16 \text{ cm}$$

$$QS = CD - CQ - DS = 18 -$$

$$\frac{1}{6} \times 18 - \frac{1}{6} \times 18 = 12 \text{ cm}$$

$$PQ = EF - QF - PE = 12 - \frac{1}{6} \times 12 - \frac{1}{6} \times 12 = 8 \text{ cm}$$

$$RS = GH - SH - GR = 30 -$$

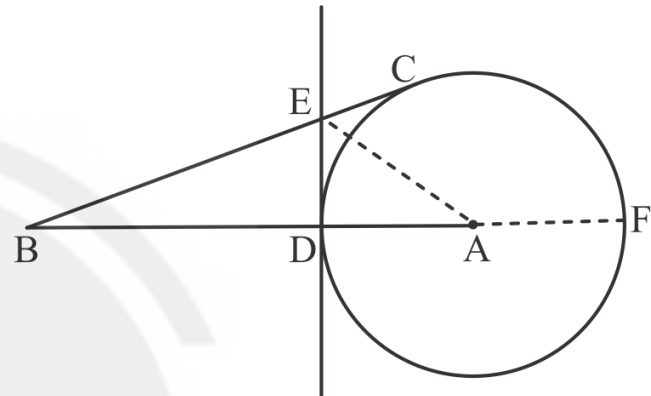
$$\frac{1}{6} \times 30 - \frac{1}{6} \times 30 = 20 \text{ cm}$$

Hence, the perimeter of the quadrilateral = $(16 + 12 + 8 + 20) = 56 \text{ cm}$

Video Solution:



Q11 Text Solution:



Let F be the point where BA intersects the circle when it is extended, so BF will be a secant to the circle, so

$$BC^2 = BD \times BF$$

$$\Rightarrow 36^2 = 24 \times BF$$

$$\Rightarrow BF = 54$$

If we consider the radius of the circle to be r then,

$$BF = BD + 2r$$

So

$$r = \frac{BF - BD}{2} = 15 \text{ cm}$$

Since both the triangles ACB and EDB are right angle triangles and have B as common angle, they are similar triangles, so

$$\frac{ED}{AC} = \frac{BD}{BC}$$

$$\Rightarrow \frac{ED}{15} = \frac{24}{36}$$

$$\Rightarrow ED = 10 \text{ cm}$$

So

$$AE =$$

$$\sqrt{AD^2 + DE^2} = \sqrt{15^2 + 10^2} = 5\sqrt{13} \text{ cm}$$

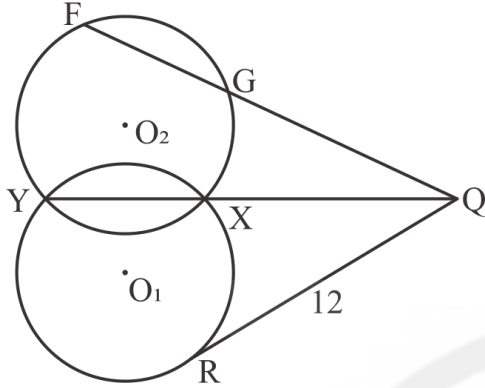
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Q12 Text Solution:


For circle with center O_1

$$QR^2 = QX \times QY$$

For circle with center O_2

$$QG \times QF = QX \times QY$$

So

$$QR^2 = QG \times QF$$

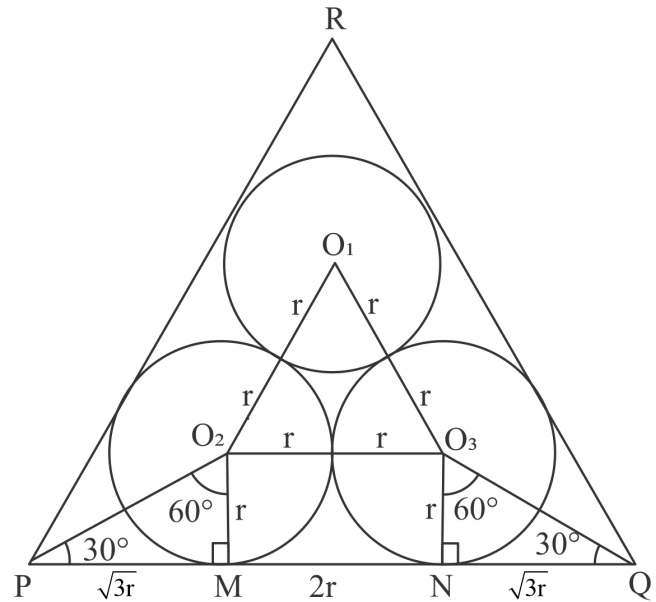
$$\Rightarrow 144 = QF \times 8$$

$$\Rightarrow QF = 18$$

So

$$FG = QF - QG = 18 - 8 = 10 \text{ cm}$$

Video Solution:

Q13 Text Solution:


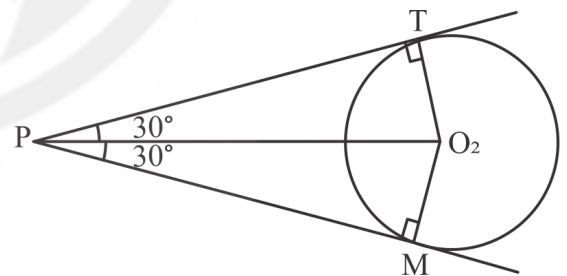
Let the three centers be O_1, O_2 and O_3

If we take the radius of the circles to be r then, the sides of triangle formed by joining the three centers will be $2r$ each and it will be an equilateral triangle.

Joining P and O_2 and joining O_2 to M (Point where PQ is tangential to the circle) we obtain right angle triangle O_2MP with M as right angle

$$\text{Angle } O_2PM = \frac{1}{2}(\angle RPQ) = 30^\circ$$

(Because through P there are two tangents to the circle forming two right angle triangles)



The two radii are equal, hypotenuse of both triangles are equal and hence by RHS congruency they are congruent and hence P O_2 is bisecting angle P

So the triangle will become a $30:60:90$ triangle giving us

$$PM = \sqrt{3}r \text{ cm}$$

$$\text{Similarly } NQ = \sqrt{3}r \text{ cm}$$

$$\text{Also } MN = 2r$$

So



$$PQ = PM + MN + NQ =$$

$$\sqrt{3}r + 2r + \sqrt{3}r = 2(1 + \sqrt{3})r \text{ cm}$$

Since PQR is also an equilateral triangle, the area of triangle PQR and that of triangle

$O_1O_2O_3$ will be in the ratio of square of their sides, so

$$\frac{ar(\triangle O_1O_2O_3)}{ar(\triangle PQR)} = \frac{(O_2O_3)^2}{(PQ)^2}$$

$$\Rightarrow ar(\triangle O_1O_2O_3) = \frac{(2r)^2}{[2(1+\sqrt{3})r]^2}$$

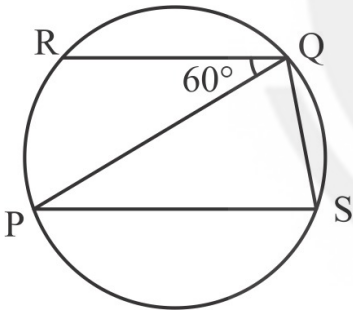
$$\times (32 + 16\sqrt{3}) = \frac{1}{4+2\sqrt{3}} \times (32 + 16\sqrt{3})$$

$$= 8 \text{ cm}^2$$

Video Solution:



Q14 Text Solution:



If PQ are as far as possible then PQ must be diameter of the circle. and from the circle property we know that the angle made by diameter is always 90 degree.

hence triangle PRQ will be a right angled triangle.

From 30-60-90 triangle property, PQ will be twice of QR

$$\text{So, } PQ = 2(QR) = 50 \text{ cm}$$

Therefore triangle PQS will also be a right angled triangle with the same property.

So using Pythagoras, PS =

$$\sqrt{50^2 - 14^2} = 48$$

Hence the area of triangle PQS =

$$\frac{1}{2} \times 14 \times 48 = 336 \text{ cm}^2$$

Video Solution:



Q15 Text Solution:

Since, the measure of the central angle is 20% of the complete angle,

i.e., the ratio, central angle : complete angle = 1 : 5

Now, given that the measure of the arc length is 44 cm.

Let the radius of the circle be r cm.

Hence, the length of the arc =

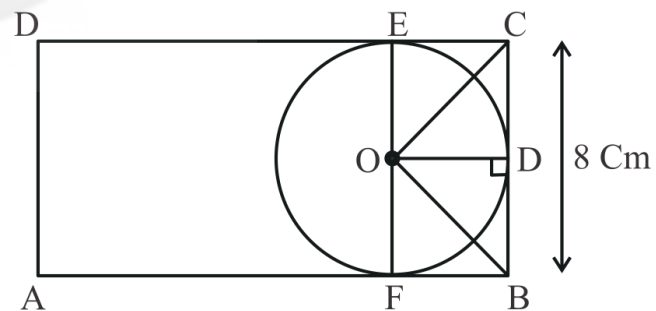
$$\frac{1}{5} \times 2 \times \frac{22}{7} \times r = 44 \text{ (given)}$$

$$r = 35 \text{ cm.}$$

Video Solution:



Q16 Text Solution:



Since, the area of the circle is $16\pi \text{ cm}^2$, so the radius of the circle is 4 cm.

According to the given condition, the diameter of the circle EF = The length of BC = $2 \times 4 = 8$ cm [The radius of the circle OF is perpendicular to the side AB and $BC \perp AB$,

so EF || BC and since, AB = CD, so EF = BC]



Again, $OD \perp BC$. So, from the right-angled triangle ODB, using the Pythagorean Theorem, we get

$$OB = \sqrt{OD^2 + BD^2} = \sqrt{4^2 + 4^2} \text{ cm.} \\ = 4\sqrt{2}$$

[Since, O is the mid-point of EF, so D is the mid-point of BC as $EF = BC$ and $OD \parallel EC$]

Similarly, $OC = 4\sqrt{2}$ cm.

Thus, the perimeter of the triangle OBC = $4\sqrt{2} + 4\sqrt{2} + 8 = 8(1 + \sqrt{2})$ cm

Video Solution:

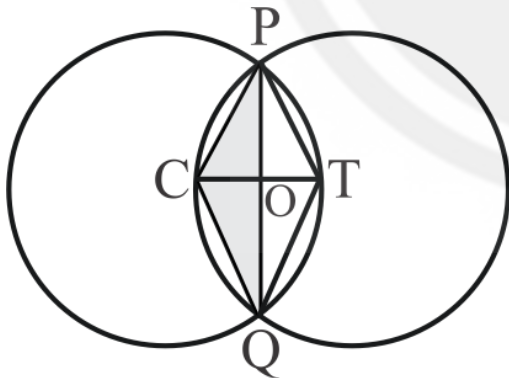


Q17 Text Solution:

According to the given figure, the radius of each circle is the same. Given that, the area of the circle with centre C is $144\pi \text{ cm}^2$.

So, the radius = 12 cm.

Let's join the lines PT and QT. Also, join the line CT.



Observe that, the diamond shape is divided into two equal parts of equilateral triangles PCT and CQT (since, $PC = PT = CT = CQ = QT = \text{radius}$).

Now, in the equilateral triangle PCT, PO is the median, so the area of the triangle PCO = $\frac{1}{2} \times$ area of triangle PCT

$$= \frac{1}{2} \times \frac{\sqrt{3}}{4} \times 12^2 = 18\sqrt{3} \text{ cm}^2$$

Similarly, the area of the triangle CQO =

$$18\sqrt{3} \text{ cm}^2$$

Hence, the area of the shaded region =

$$2 \times 18\sqrt{3} = 36\sqrt{3} \text{ cm}^2$$

Video Solution:

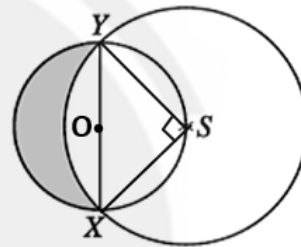


Q18 Text Solution:

Let the radius of the smaller circle = $OS = OY =$

$OX = r$ cm (radius of the smaller circle)

Let the radius of the larger circle = x cm.



Therefore, $\angle YSX = 90^\circ$

Now, from the right-angled triangle

$$YSX, x^2 + x^2 = (2r)^2$$

$$r^2 = \frac{x^2}{2} \dots (i)$$

So, the area of the sector YX in the larger circle

$$= \frac{90}{360} \times \pi \times x^2 - \frac{1}{2} \times x^2 \times \sin 90^\circ = \frac{\pi x^2}{4} - \frac{x^2}{2}$$

Thus, the area of the shaded region = The area of the semicircle of the smaller circle - the area of the sector YX

$$= \frac{\pi r^2}{2} - \frac{\pi x^2}{4} + \frac{x^2}{2}$$

$$= \frac{\pi}{2} \times \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{x^2}{2} \quad [\text{Using (i)}]$$

$$= \frac{x^2}{2} = 8 \text{ cm}^2 \quad (\text{given})$$

$$x = 4 \text{ cm}$$

Hence, the area of the larger circle = $16\pi \text{ cm}^2$

Video Solution:



**Q19 Text Solution:**

Given that, the perimeter of the shaded region MNQP is given as 20 cm.

Therefore, we have: $MN + NQ + PQ + PM = 20$
i.e., $MN + PQ + 2NQ = 20$ [Since, $NQ = PM$ as the difference between the radii of the sectors]

i.e., $x + 2NQ = 20$ [Let $MN + PQ = x$]

So, $NQ = \frac{20-x}{2}$. i.e., the maximum integer value of NQ is 9 cm.

Thus, $OQ = ON + NQ = 3 + 9 = 12$ cm.

Let $\angle POQ = \theta$, then $MN = \frac{\theta}{360^\circ} \times 2 \times \pi \times 3 = 3\theta$ (in radian)

Similarly, $PQ = \frac{\theta}{360^\circ} \times 2 \times \pi \times 12 = 12\theta$ (in radian)

Now, $MN + PQ + PM + NQ = 20$

$$3\theta + 12\theta + 9 + 9 = 20$$

$$15\theta + 9 \times 2 = 20$$

$$15\theta = 2$$

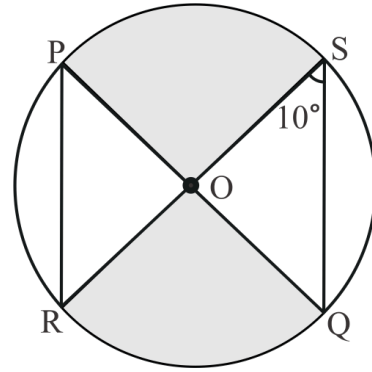
$$\theta = \frac{2}{15}$$

Hence, the required area of the region = Area of the region OQPO - Area of ONMO

=

$$\frac{\pi}{360} \times \frac{2}{15} \times \{OQ^2 - ON^2\}$$

$$= \frac{1}{2} \times \frac{2}{15} \times \{OQ^2 - ON^2\} = \frac{1}{2} \times \frac{2}{15} \times \{12^2 - 3^2\} = 9 \text{ cm}^2$$

Video Solution:**Q20 Text Solution:**

PQ and RS are diameters of circle. Intersecting point of two diameters is a centre of circle.

Let the point of intersection. i.e., centre be 'O'.

$RO = OS = \text{radius of circle} = 42$ cm.

$PO = OQ = 42$ cm.

$$\angle RSQ = 10^\circ = \angle SQP$$

$$\angle QOS = 180^\circ - 10^\circ - 10^\circ = 160^\circ$$

$$\angle ROQ = 180^\circ - 160^\circ = 20^\circ$$

Similarly,

$$\angle POR = 160^\circ \text{ and } \angle POS = 20^\circ$$

Area of the shaded region

=

$$2 \times \frac{20}{360} \times \pi \times (42)^2 = \frac{1}{9} \times \frac{22}{7} \times 42 \times 42 = 616 \text{ cm}^2$$

Video Solution:**Q21 Text Solution:**

Given that the area of the A-centric circle is:

$$(3 + 2\sqrt{2})\pi$$

$$= \left[(1)^2 + 2 \times 1 \times \sqrt{2} + (\sqrt{2})^2 \right] \pi$$

$$= (\sqrt{2} + 1)^2 \pi \text{ cm}^2$$

So, the radius of the circle = $\sqrt{2} + 1$ cm.

Now, by the given diagram, we have

$$AB = AE = AD = \text{radius} = \sqrt{2} + 1 \text{ cm}$$

Now, since, ABCD is a square, so $BC \perp AB$.

Let $EC = x$ cm.



Then, from the right-angled triangle ABC, we get

$$AC^2 = AB^2 + BC^2$$

$$(\sqrt{2} + 1 + x)^2 = 2(\sqrt{2} + 1)^2$$

$$\sqrt{2} + 1 + x = \pm\sqrt{2}(\sqrt{2} + 1)$$

$$x = 2 + \sqrt{2} - \sqrt{2} - 1 \quad \text{or} \quad x = -2$$

$$-\sqrt{2} - \sqrt{2} - 1 \quad (\text{This is not possible})$$

$$x = 1 \text{ cm}$$

So, the length of EC = 1 cm.

Alternatively,

Given that the area of the A-centric circle is:

$$(3 + 2\sqrt{2})\pi$$

$$= [(1)^2 + 2 \times 1 \times \sqrt{2} + (\sqrt{2})^2]\pi$$

$$= (\sqrt{2} + 1)^2 \pi \text{ cm}^2$$

So, the radius of the circle, AB = AE = AD =

$$\sqrt{2} + 1 \text{ cm.}$$

Since, ABCD is a square, so its length of

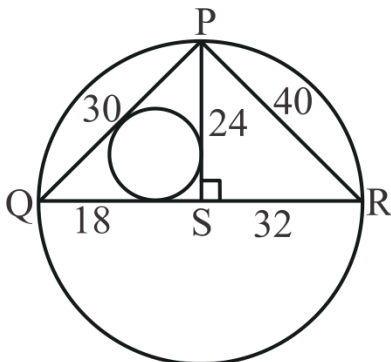
$$\text{diagonal AC} = \sqrt{2} \times (\sqrt{2} + 1) = 2 + \sqrt{2} \text{ cm.}$$

$$\text{So, EC} = \text{AC} - \text{AE} = 2 + \sqrt{2} - \sqrt{2} - 1 = 1 \text{ cm.}$$

Video Solution:



Q22 Text Solution:



Using the Pythagoras Theorem, from triangle PQR, we get

$$QR = \sqrt{PQ^2 + PR^2} = \sqrt{30^2 + 40^2} = 50 \text{ cm}$$

Now, the area of the triangle PQR,

$$\frac{1}{2} \times PQ \times PR = \frac{1}{2} \times QR \times PS$$

$$PS = \frac{PQ \times PR}{QR} = \frac{30 \times 40}{50} = 24 \text{ cm}$$

Now, again, by using the Pythagoras theorem on triangle PQS, we get

$$QS = \sqrt{PQ^2 - PS^2} = \sqrt{30^2 - 24^2} \text{ cm} = 18$$

So, inradius of the triangle PQS =

$$\frac{\text{Area}}{\text{semi-perimeter}} = \frac{\frac{1}{2} \times 18 \times 24}{\frac{30+18+24}{2}} = 6 \text{ cm}$$

Now, the circumradius of the right-angled

$$\text{triangle PQR} = \frac{\text{hypotenuse}}{2} = \frac{50}{2} = 25 \text{ cm}$$

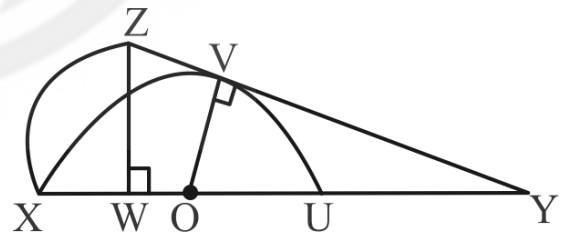
Hence, the required difference =

$$\pi [(25)^2 - 6^2] = 589\pi$$

Video Solution:



Q23 Text Solution:



Let's draw a perpendicular line ZW from Z to XY. Also, connect the radius OV of the semicircle.

Since, the area of the semicircle is given as $\frac{9\pi}{2}$,

so the radius of the semicircle $OV = \frac{3}{2} \text{ cm}$

Also, since, $YX = YZ = \text{radius of the sector} = 6 \text{ cm}$ and $OU = OV = \frac{3}{2} \text{ cm}$, so

$$YO = (XY - XU) + OU = 6 - 3 + \frac{3}{2} = \frac{9}{2} \text{ cm}$$



Now, $\triangle WYZ \sim \triangle VYO$ [Since, $\angle W = \angle V$, $\angle Y = \angle Y$]

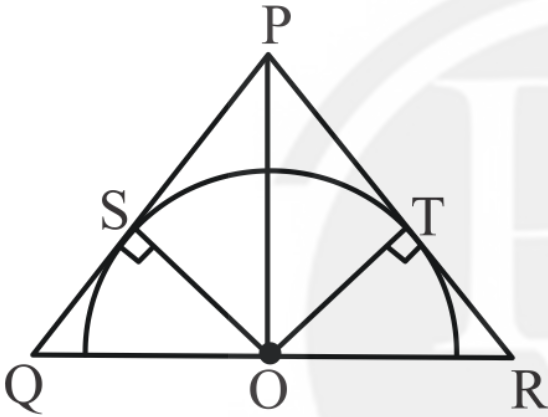
Therefore, we have: $\frac{WZ}{YZ} = \frac{VO}{YO}$

$$\Rightarrow \frac{WZ}{6} = \frac{\frac{3}{2}}{\frac{9}{2}} \Rightarrow WZ = 2 \text{ cm}$$

Video Solution:



Q24 Text Solution:



Let's join the line P, O.

Now, PQ and PR are two tangents to the semicircle at S and T respectively. So,

$OS \perp PQ$ and $OT \perp PR$.

Let the radius of the semicircle = r cm.

So, the area of the triangle PQR = area of triangle PQO + area of triangle PRO

$$= \frac{1}{2} \times PQ \times SO + \frac{1}{2} \times PR \times OT \text{ cm}^2 \\ = \frac{r}{2} (16 + 30) = 23r$$

Now, the semiperimeter of the triangle PQR =

$$\frac{16+34+30}{2} = 40$$

So, the area of the triangle PQR =

$$\sqrt{40(40-16)(40-34)(40-30)} = 240 \text{ cm}^2$$

Therefore, we have: $23r = 240$

$$r = \frac{240}{23}$$

Hence, the required ratio, area of semicircle : area of the triangle PQR

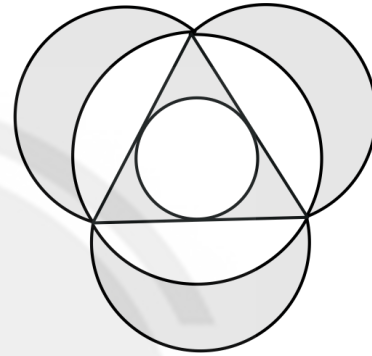
=

$$\frac{1}{2} \times \pi \left(\frac{240}{23} \right)^2 : 240 = \pi \times \frac{120}{529} : 1 = 120\pi : 529$$

Video Solution:



Q25 Text Solution:



We are given an equilateral triangle with side length of 12 cm.

Circumradius of the triangle =

$$\frac{1}{\sqrt{3}} \times 12 = 4\sqrt{3} \text{ cm}$$

$$\text{Inradius for the triangle} = \frac{1}{2\sqrt{3}} \times 12 = 2\sqrt{3} \text{ cm}$$

Also, the area of the triangle =

$$\frac{\sqrt{3}}{4} \times (12)^2 = 36\sqrt{3} \text{ cm}^2$$

Area of each semicircle =

$$\frac{1}{2} \times \pi \times (6)^2 = 18\pi \text{ cm}^2$$

Thus, the area of the outer circle =

$$\pi \times (4\sqrt{3})^2 = 48\pi \text{ cm}^2$$

Area of the inner circle =

$$\pi \times (2\sqrt{3})^2 = 12\pi \text{ cm}^2$$

So, the area of the shaded region = [3(area of the semicircle) + area of the triangle] - area of the outer circle + (area of the triangle - area of the inner circle)

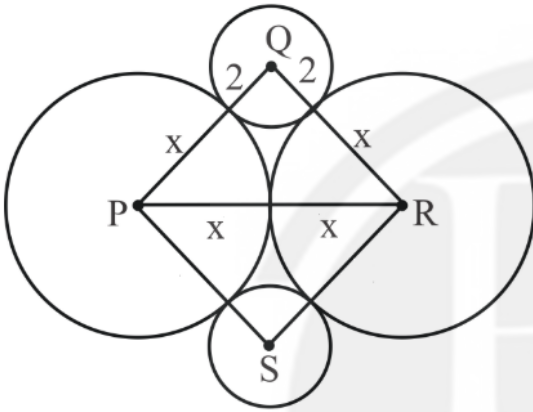


$$\begin{aligned}
 & [(3 \times 18\pi + 36\sqrt{3}) - 48\pi] \\
 & + [36\sqrt{3} - 12\pi] \\
 & = 72\sqrt{3} - 6\pi \text{ cm}^2
 \end{aligned}$$

Video Solution:



Q26 Text Solution:



In a triangle PQR, $\angle Q = 90^\circ$

Using the Pythagoras Theorem,

$$(x+x)^2 = (x+2)^2 + (x+2)^2$$

$$4x^2 = 2(x+2)^2$$

$$2x^2 = x^2 + 4x + 4$$

$$x^2 - 4x + 4 = 8$$

$$(x-2)^2 = 8$$

$$x-2 = 2\sqrt{2}$$

$$x = 2(\sqrt{2} + 1)$$

Thus, the side of square = $x+2 =$

$$2(\sqrt{2} + 1) + 2 = 2\sqrt{2} + 4 \text{ cm}$$

So, the area of the square

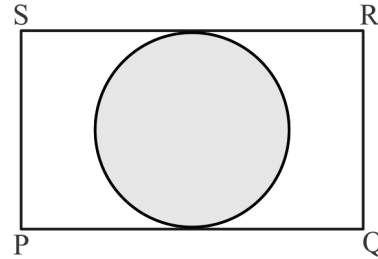
$$= (2\sqrt{2} + 4)^2 = 8(1 + \sqrt{2})^2$$

$$= 8(3 + 2\sqrt{2}) \text{ cm}^2$$

Video Solution:



Q27 Text Solution:



Let the length and breadth of the rectangle be $2l$ and $2b$, respectively.

Therefore, the area of the circle = πb^2 .

Now, according to the given condition, we get

$$\frac{1}{3} \times (2l \times 2b - \pi b^2) = \pi b^2$$

$$4lb = 4\pi b^2$$

$$l = \pi b$$

Given that,

$$4lb = 81$$

$$4 \times \pi b^2 = 81$$

$$b = \frac{9}{2\sqrt{\pi}} \Rightarrow l = \pi b = \frac{9\sqrt{\pi}}{2}$$

Hence, the required perimeter of the rectangle

$$= 2(2l + 2b) = 4(l + b) = 4\left(\frac{9\sqrt{\pi}}{2} + \frac{9}{2\sqrt{\pi}}\right)$$

$$= 2 \times 9\sqrt{\pi}\left(1 + \frac{1}{\pi}\right)$$

Video Solution:



Q28 Text Solution:

Since, OABCO is a quadrant of a circle, so

$$\angle COA = 90^\circ.$$

Now, according to the given diagram, AC must be the diameter of the circle with radius $14\sqrt{2}$ cm, since, this circle is the circumcircle of the right-angled triangle COA.



$$\text{Thus, } OC^2 + OA^2 = AC^2$$

$$2OA^2 = (2 \times 14\sqrt{2})^2$$

$$[\text{Since, } OA = OC]$$

$$OA = \frac{28\sqrt{2}}{\sqrt{2}} = 28 \text{ cm}$$

Hence, the area of the shaded region = area of the quadrant - area of the triangle COA

$$= \frac{1}{4} \times \frac{22}{7} \times 28^2 - \frac{1}{2} \times 28^2$$

$$= 224 \text{ cm}^2$$

Now, the area of the leftover portion of the circle = area of the circle - area of the shaded portion

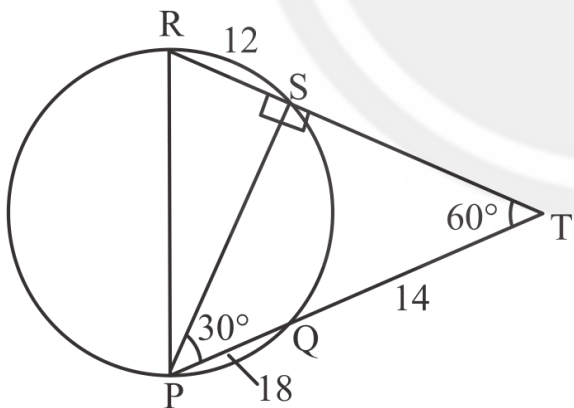
$$= \frac{22}{7} \times (14\sqrt{2})^2 - 224$$

$$= 1008 \text{ cm}^2$$

Video Solution:



Q29 Text Solution:



Using the intersecting secant theorem and taking $ST = a$ cm

$$(RS + ST)(ST) = (PQ + QT)(QT)$$

$$\Rightarrow (12 + a)(a) = (18 + 14)(14)$$

$$\Rightarrow a^2 + 12a = 448$$

$$\Rightarrow a^2 + 12a - 448 = 0$$

$$\Rightarrow a^2 + 28a - 16a - 448 = 0$$

$$\Rightarrow a(a + 28) - 16(a + 28) = 0$$

$$\Rightarrow a = 16 \text{ or } -28$$

Since the length cannot be negative

$$ST = 16 \text{ cm}$$

Looking at the triangle PST

$$ST = 16 \text{ cm}$$

$$PT = 32 \text{ cm (ST:PT = 1:2)}$$

$$\angle PTS = 60^\circ$$

Which means that it is a 30:60:90 triangle and angle PST is a right angle, so

$$PS = \sqrt{3}ST = 16\sqrt{3} \text{ cm}$$

Since angle RSP is a right angle formed by PR on the circumference, PR will be the diameter of the circle, so

$$PR = \sqrt{RS^2 + PS^2} = \sqrt{12^2 + (16\sqrt{3})^2}$$

$$= \sqrt{912} \text{ cm}$$

So area of the circle =

$$\pi \left(\frac{\text{Diameter}}{2} \right)^2 = \pi \left(\frac{\sqrt{912}}{2} \right)^2 = \frac{912}{4} \pi$$

$$= 228\pi \text{ cm}^2$$

Video Solution:

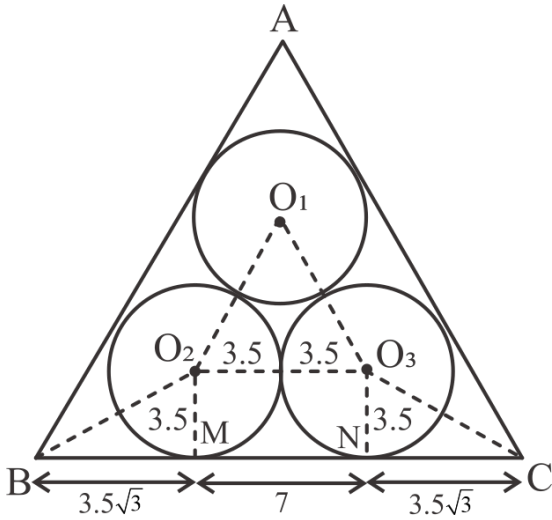


Q30 Text Solution:

Let O_1 , O_2 and O_3 be the centers of the, *three circles*

and let ABC be the triangle formed by the three tangents. The triangle ABC will be an equilateral triangle as the radii of all the three circles are equal and in such cases whenever tangential sides common to two triangles are taken, they form an equilateral triangle.





So

$$AB = BC = AC$$

Also

$$\angle A = \angle B = \angle C = 60^\circ$$

So the line segments joining the vertices of the triangles to the center of the triangle will bisect them into two angles of 30° .

Let M and N be the point of contact of the tangent BC to the circles with center

O_2 and O_3 .

So

$$O_2M = O_3M = \text{radius}(r) = 3.5 \text{ cm}$$

Also in triangle BMO_2 ,

$$\angle O_2MB = 90^\circ \text{ (Perpendicular)}$$

$$\angle O_2BM = 30^\circ$$

and hence

$$\angle BO_2M = 60^\circ$$

So the triangle is a 30:60:90 triangle and hence

the sides will be in ratio $1 : \sqrt{3} : 2$

So

$$BM = 3.5\sqrt{3} \text{ cm}$$

Similarly

$$CN = 3.5\sqrt{3} \text{ cm}$$

So

$$BC = BM + MN + CN =$$

$$3.5\sqrt{3} + 7 + 3.5\sqrt{3} = 7(1 + \sqrt{3}) \text{ cm.}$$

$$(MN = O_2O_3 = 2r = 7 \text{ cm})$$

So the area of the equilateral triangle =

$$\frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}[7(1 + \sqrt{3})]^2 \text{ cm}^2$$

Area of the three circles =

$$3 \times \pi r^2 = 3\left(\frac{22}{7}\right)(3.5)^2 \text{ cm}^2$$

So the area of the triangle that is not covered by the circles

= *area of the triangle* – *area of the circles*

$$= \frac{\sqrt{3}}{4}[7(1 + \sqrt{3})]^2 - 3\left(\frac{22}{7}\right)(3.5)^2$$

$$= \frac{\sqrt{3}}{4}(7)^2(4 + 2\sqrt{3}) - 3\left(\frac{22}{7}\right)(3.5)^2$$

$$= \sqrt{3}(3.5)^2\left(4 + 2\sqrt{3} - \frac{22\sqrt{3}}{7}\right)$$

$$= \frac{\sqrt{3}(3.5)^2(28 - 8\sqrt{3})}{7}$$

$$= \sqrt{3}(3.5)\left(14 - 4\sqrt{3}\right)$$

$$= \left(49\sqrt{3} - 42\right) \text{ cm}^2$$

Video Solution:



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